

Erdős Problem #706

EP-706

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $L(r)$ be such that if G is a graph formed by taking a finite set of points P in $\mathbb{R}^2$ and some set $A \subset (0, \infty)$ of size r , where the vertex set is P and there is an edge between two points if and only if their distance is a member of A , then $\chi(G) \leq L(r)$.

Estimate $L(r)$. In particular, is it true that $L(r) \leq r^{O(1)}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>